MH1300 FOUNDATIONS OF MATHEMATICS

Tutorial 9 Solutions

Ex. 5.4.8. Suppose that $h_0, h_1, h_2, \cdots$ is a sequence defined as follows:

$$h_0 = 1, \; h_1 = 2, \; h_2 = 3,$$
$$h_k = h_{k-1} + h_{k-2} + h_{k-3}, \text{ for all integers } k \geq 3.$$

(a) Prove that $h_n \leq 3^n$ for all $n \geq 0$.
(b) Suppose that s is any real number such that $s^3 \geq s^2 + s + 1$. (This implies that $s > 1.83$.) Prove that $h_n \leq s^n$ for all $n \geq 2$.

SOLUTION . (a) **Proof by strong mathematical induction:**
Let the property $P(n)$ be the sentence $h_n \leq 3^n$. We prove by strong mathematical induction that this property is true for all integers $n \geq 0$. Let $a = 0$ and $b = 2$.

Basis step:

When $n = 0, 1$ or 2, the sentences are $h_0 \leq 3^0, h_1 \leq 3^1$ and $h_2 \leq 3^2$ respectively. This is true since $h_0 = 1, h_1 = 2$ and $h_2 = 3$.

Inductive step:

Suppose for some integer $k \geq b = 2$, the property is true for all integers i with $0 \leq i \leq k$. In other words, for all integers i with $0 \leq i \leq k$, we know that $h_i \leq 3^i$.

Since $k \geq 2$, hence $k + 1 \geq 3$. Therefore, $h_{k+1} = h_k + h_{k-1} + h_{k-2}$. We have $0 \leq k - 2 < k - 1 < k$, this means that $P(k-2), P(k-1)$ and $P(k)$ are all true. Thus, $h_{k-2} \leq 3^{k-2}, h_{k-1} \leq 3^{k-1}$ and $h_k \leq 3^k$. Therefore,

$$\begin{aligned} h_{k+1} &= h_k + h_{k-1} + h_{k-2} \\ &\leq 3^k + 3^{k-1} + 3^{k-2} \\ &< 3^k + 3^k + 3^k \\ &= 3(3^k) = 3^{k+1}. \end{aligned}$$

Therefore, by strong mathematical induction, $P(n)$ is true for all integers $n \geq 0$.

(b) **Proof by strong mathematical induction:**
Let the property $P(n)$ be the sentence $h_n \leq s^n$, where s is a fixed real number such that $s^3 \geq s^2 + s + 1$. We prove by strong mathematical induction that this property is true for all integers $n \geq 2$. Let $a = 2$ and $b = 4$.

Basis step:

When $n = 2$, the sentence is $h_2 \leq s^2$. Since $h_2 = 3$, we have $h_2 = 3 < 1.83^2 < s^2$. Thus, $P(2)$ is true.
When $n = 3$, the sentence is $h_3 \leq s^3$. Since $h_3 = 6$, we have $h_3 = 6 < 1.83^3 < s^3$. Thus, $P(3)$ is true.
When $n = 4$, the sentence is $h_4 \leq s^4$. Since $h_4 = 11$, we have $h_4 = 11 < 1.83^4 < s^4$. Thus, $P(4)$ is true.

Inductive step:

Suppose for some integer $k \geq b = 4$, the property is true for all integers i with $a = 2 \leq i \leq k$. In other words, for all integers i with $2 \leq i \leq k$, we know that $h_i \leq s^i$.

Since $k \geq 4$, hence $k+1 \geq 3$. Therefore, $h_{k+1} = h_k + h_{k-1} + h_{k-2}$. We have $2 \leq k-2 < k-1 < k$, this means that $P(k-2), P(k-1)$ and $P(k)$ are all true. Thus, $h_{k-2} \leq s^{k-2}, h_{k-1} \leq s^{k-1}$ and $h_k \leq s^k$. Therefore,

$$\begin{aligned} h_{k+1} &= h_k + h_{k-1} + h_{k-2} \\ &\leq s^k + s^{k-1} + s^{k-2} \\ &= s^{k-2}(s^2 + s + 1) \\ &\leq s^{k-2}(s^3) \\ &= s^{k+1}. \end{aligned}$$

Therefore, by strong mathematical induction, $P(n)$ is true for all integers $n \geq 0$. □

Ex. 5.4.13. Use Strong mathematical induction to prove the existence part of the unique factorization theorem (Theorem 4.3.5): Every integer greater than 1 is either a prime number or a product of prime numbers. (Uniqueness part is in Ex. 8.4.41.)

SOLUTION . **Proof by strong mathematical induction:**

Let the property $P(n)$ be the sentence

"n is either a prime number or a product of prime numbers."

We prove by strong mathematical induction that this property is true for all integers $n \geq 2$.

Basis step:

When $n = 2$, the sentence is "2 is either a prime number or a product of prime numbers." This sentence is true because 2 is prime.

Inductive step:

Suppose for some integer $k \geq 2$, the property is true for all integers i with $2 \leq i \leq k$. In other words, for all integers i with $2 \leq i \leq k$, we know that i is either a prime number or a product of prime numbers. [This is the inductive hypothesis.]

We must show that $k+1$ is either a prime number or a product of prime numbers. [i.e. W.T.S. that $P(n)$ is true for $n = k+1$.]

In case $k+1$ is prime, the sentence is true. So suppose $k+1$ is not prime. Then $k+1$ is composite and by the definition of composite, $k+1 = rs$ for some positive integers r and s with $1 < r < k+1$ and $1 < s < k+1$.

Thus, by inductive hypothesis, each of r and s is either a prime number or a product of prime numbers. So, because $k+1$ is the product of r and s, $k+1$ is a product of prime numbers [as was to be shown].

Therefore, by strong mathematical induction, $P(n)$ is true for all integers $n \geq 2$. □

Ex. 5.4.18. Compute $9^0, 9^1, 9^2, 9^3, 9^4$ and 9^5. Make a conjecture about the units digit of 9^n where n is a positive integer. Use (strong) mathematical induction to prove your conjecture.

SOLUTION .

$$9^0 = 1, \qquad 9^1 = 9, \qquad 9^2 = 81, \qquad 9^3 = 729, \qquad 9^4 = 6561, \qquad 9^5 = 59049.$$

Conjecture: For any positive integer n, the units digit of 9^n is 9 if n is odd and it is 1 if n is even.

Proof by strong mathematical induction:

Let the property $P(n)$ be the sentence

"the units digit of 9^n is 9 if n is odd and it is 1 if n is even."

We prove by strong mathematical induction that this property is true for all integers $n \geq 1$.

Basis step:

When $n = 1$, we see that 1 is odd and $9^1 = 9$ has units digit 9. When $n = 2$, we see that 2 is even and $9^2 = 81$ has units digit 1. Hence both $P(1)$ and $P(2)$ are true.

Inductive step:

Suppose for some integer $k \geq 2$, the property is true for all integers i with $1 \leq i \leq k$. In other words, for all integers i with $1 \leq i \leq k$, we know that the units digit of 9^i is 9 if i is odd and it is 1 if i is even. [This is the inductive hypothesis.]

We must show that the units digit of 9^{k+1} is 9 if $k+1$ is odd and it is 1 if $k+1$ is even. [i.e. W.T.S. that $P(n)$ is true for $n = k+1$.]

Case 1: $k+1$ is odd.

Then k is even, and by the induction hypothesis, the units digit of 9^k is 1. Therefore, for some integer m, we have $9^k = 10m + 1$. Therefore,

$$9^{k+1} = 9 \cdot 9^k = 9(10m+1) = 10 \cdot (9m) + 9.$$

Since $9m$ is an integer, the units digit of 9^{k+1} is 9.

Case 2: $k+1$ is even.

Then k is odd, and by the induction hypothesis, the units digit of 9^k is 9. Therefore, for some integer m, we have $9^k = 10m + 9$. Therefore,

$$9^{k+1} = 9 \cdot 9^k = 9(10m+9) = 10 \cdot (9m) + 81 = 10 \cdot (9m+8) + 1.$$

Since $9m + 8$ is an integer, the units digit of 9^{k+1} is 1.

Therefore, in both cases $P(k+1)$ is true.

Therefore, by strong mathematical induction, $P(n)$ is true for all integers $n \geq 1$.

Remark: Notice that this solution is considered a "strong mathematical induction" because there are two base cases $P(1)$ and $P(2)$. This is actually unnecessary and it is actually fine to only consider $P(1)$ as the basis step. Thus, we could prove this by regular mathematical induction. $\square$

Ex. 5.4.26. Suppose $P(n)$ is a property such that

(1) $P(0), P(1)$ and $P(2)$ are all true.

(2) for all integers $k \geq 0$, if $P(k)$ is true, then $P(3k)$ is true. Must it follow that $P(n)$ is true for all integers $n \geq 0$? If yes, explain why; if no, give a counterexample.

Additional: If the answer to (2) above is no, make one *small* change so that $P(n)$ is true for all $n \geq 0$.

SOLUTION . This is false, because how can you conclude that $P(4)$ is true? The items (1) and (2) will only guarantee that $P(3^k)$ and $P(2 \cdot 3^k)$ are true for every $k \geq 0$, but do not guarantee the other cases. For example, we can take $P(n)$ to be the statement "Either $n < 3$ or n is divisible by 3". Then, $P(0), P(1)$ and $P(2)$ are obviously true. For any k, $P(3k)$ is true since $3k$ is divisible by 3. However, $P(n)$ is not true for all integers $n \geq 0$, because for instance, $P(4)$ is not true, since 4 is neither less than nor divisible by 3.

To modify the statement so that $P(n)$ is true for all n, we change condition (2) to condition (2'): "for all integers $k \geq 0$, if $P(k)$ is true, then $P(k+3)$ is true." Then $P(n)$ will be true for all $n \geq 0$. To prove that $P(n)$ is true for all n, we proceed by strong induction with $a = 0, b = 2$. Assumption (1) says that $P(0), P(1)$ and $P(2)$ are true and will give us the basis step. Now suppose $k \geq 2$ and suppose that $P(0), P(1), \cdots, P(k)$ are all true. Since $k \geq 2$, this means that $k - 2 \geq 0$. Thus, $P(k-2)$ is true. Therefore, by our assumption (2'), since $P(k-2)$ is true, we conclude that $P((k-2)+3) = P(k+1)$ is true. Thus, by strong Mathematical Induction, $P(n)$ is true for all $n \geq 0$. $\square$

Ex. E1 (The tower of Hanoi) The "Tower of Hanoi" is a well-known mathematical puzzle where we have three rods and n disks of different diameters. At the beginning of the game all the n disks are stacked at the first rod, in order of size, with the largest disk being at the bottom of the stack. The objective of the puzzle is to move the entire stack to the third rod, obeying the following simple rules:

Prove that for each $n \geq 1$, we can solve the puzzle using $2^n - 1$ moves.

SOLUTION . **Proof by mathematical induction:**

Let the property $P(n)$ be the sentence "The Tower of Hanoi puzzle with n disks can be solved in $2^n - 1$ moves". We prove by mathematical induction that this property is true for all integers $n \geq 1$.

Basis step:

When $n = 1$, the puzzle has only 1 disk. We can solve it in 1 move by moving the single disk to the third rod.

Inductive step:

Suppose for some integer $k \geq 1$, the property is true for k. In other words, we know that the puzzle with k disks has a solution with $2^k - 1$ many moves. Notice that by reversing the roles of rods 2 and 3, we can also solve each puzzle by requiring that we move all the disks to rod 2 instead of rod 3.

Now consider the game with $k + 1$ disks. By assuming that the bottom most disk is never moved, we can consider the problem of moving the first k many disks from the first rod to the second rod. By the inductive hypothesis $P(k)$, there is a solution to this using $2^k - 1$ many moves. (Since the bottom most disk is the largest disk and is never moved, we can stack any of the other disks on top of it, so it does not cause any moves in the solution to become an illegal move). Now after moving the first k disks from the first rod to the second rod, we move the largest disk from the first rod to the third rod, which takes one move. After this, we apply the inductive hypothesis again to move the k disks from rod 2 to rod 3. This takes a further $2^k - 1$ moves. Thus, the total number of moves is $(2^k - 1) + 1 + (2^k - 1) = 2^{k+1} + 1$.

Therefore, by mathematical induction, $P(n)$ is true for all integers $n \geq 0$. $\square$

Ex. 4.8.16. Use the Euclidean algorithm to hand-calculate the greatest common divisors of the pair of integers

$$4131 \quad \text{and} \quad 2431.$$

SOLUTION .

$$\begin{aligned} 4131 &= 2431 \cdot 1 \ + \ 1700 \\ 2431 &= 1700 \cdot 1 \ + \ 731 \\ 1700 &= 731 \cdot 2 \ + \ 238 \\ 731 &= 238 \cdot 3 \ + \ 17 \\ 238 &= 17 \cdot 14 \ + \ 0. \end{aligned}$$

Therefore $\gcd(4131, 2431) = 17$. $\square$

Ex. 4.8.21. Complete the proof of Lemma 4.8.2 by proving the following: If a and b are any integers with $b \neq 0$ and q and r are any integers such that

$$a = bq + r,$$

then

$$\gcd(b, r) \leq \gcd(a, b).$$

SOLUTION . 1. A common divisor of b and r is also a common divisor of a and b.
Let a and b be integers with $b \neq 0$ and q and r are any integers such that $a = bq + r$. Suppose $c \in \mathbb{Z}$ is a common divisor of b and r.
Then $c|b$ and $c|r$, and so by a previous result from lecture, $c|(bx + ry)$ for any integers x and y. In particular,

$$c|(bq + r).$$

Since $a = bq + r$, this means $c|a$. As $c|b$ also, we see that c is a common divisor of a and b.

2. Conclude $\gcd(b, r) \leq \gcd(a, b)$. By Step 1, every common divisor of b and r is also a common divisor of b and r.
It follows that $\gcd(b, r)$, the greatest common divisor of b and r, is also a divisor of a and b.
Hence $\gcd(b, r)$ is less than or equal to the greatest common divisor of a and b, i.e. $\gcd(b, r) \leq \gcd(a, b)$. $\square$

Ex. 4.8.27.
Definition: The **least common multiple** of two nonzero integers a and b, denoted $\mathbf{lcm(a, b)}$, is the positive integer c such that

a. $a|c$ and $b|c$,
b. for all positive integers m, if $a|m$ and $b|m$, then $c \le m$.

Prove that for all positive integers a and b, $a|b$ if and only if $\text{lcm}(a, b) = b$.

SOLUTION . Let positive integers a and b be given.

1. Show "if $a|b$ then $\text{lcm}(a, b) = b$".
   We show that b satisfies the two conditions for least common multiple.
   It is clear that a divides b (given) and b divides itself.
   Next, suppose we are given a positive integer m such that
   $$a|m \qquad \text{and} \qquad b|m.$$
   Then we have $b \le m$ since $b|m$ and both b and m are positive. Therefore, b satisfies both conditions of the definition of lcm. Therefore, b is the lcm of a and b.
2. Show "if $b = \text{lcm}(a, b)$ then $a|b$".
   Since b is the lcm of a and b, from the definition of lcm, we have $a|\text{lcm}(a, b) = b$, i.e., $a|b$. □

Additional Exercises On Complex Numbers

Ex. 1. Prove that for two complex numbers w and z, we have the triangle inequality $|z + w| \le |z| + |w|$. Next, use it to prove $|z - w| \ge ||z| - |w||$.

SOLUTION . First, note that the square of the left side is
$$|z + w|^2 = (z + w)(\overline{z + w}) = (z + w)(\overline{z} + \overline{w}) = z\overline{z} + z\overline{w} + w\overline{z} + w\overline{w}. \tag{1}$$
The second and third term is
$$z\overline{w} + w\overline{z} = z\overline{w} + \overline{z\overline{w}} = 2\text{Re}(z\overline{w}) \le 2\,|z\overline{w}| = 2|z||w|.$$
Applying this on equation (1), we arrive at
$$|z + w|^2 \le |z|^2 + 2|z||w| + |w|^2 = (|z| + |w|)^2\,.$$
Taking square roots of both sides and noting that moduli are nonnegative, we obtain
$$|z + w| \le |z| + |w|,$$
which is the triangle inequality we want to show.

Next, we prove the second version of the triangle inequality.
Replacing z by $z - w$, we get
$$|z| \le |z - w| + |w|,$$
which, upon rearranging, we arrive at
$$|z - w| \ge |z| - |w|. \tag{3}$$
Similarly, replacing w by $w - z$ in the triangle inequality, we obtain
$$|w| \le |z| + |w - z|,$$

which, upon rearranging, we arrive at

$$|z-w| \;=\; |w-z| \;\geq\; |w|-|z|. \tag{4}$$

Combining equations (3) and (4), we arrive at

$$|z-w| \;\geq\; ||z|-|w||.$$

$\square$

Ex. 2. Find the 6th-roots of $5-i5\sqrt{3}$.

SOLUTION . Clearly $5-i5\sqrt{3}$ lies in the fourth quadrant. Therefore,

$$\arg(5-i5\sqrt{3}) \;=\; 2\pi-\tan^{-1}\left(\frac{5\sqrt{3}}{5}\right) \;=\; 2\pi-\tan^{-1}\left(\sqrt{3}\right) \;=\; 2\pi-\frac{\pi}{3} \;=\; \frac{5\pi}{3}.$$

Next, the modulus of $5-i5\sqrt{3}$ is

$$\left|5-i5\sqrt{3}\right| \;=\; \sqrt{5^2+\left(5\sqrt{3}\right)^2} \;=\; \sqrt{25+25\cdot 3} \;=\; 5\sqrt{1+3} \;=\; 10.$$

Therefore, the exponential form of $5-i5\sqrt{3}$ is

$$5-i5\sqrt{3} \;=\; 10e^{i\frac{5\pi}{3}}.$$

Therefore, one of the 6th-roots of $5-i5\sqrt{3}$ is

$$z_0 \;=\; 10^{\frac{1}{6}}e^{i\frac{5\pi}{18}},$$

and the rest are obtained by multiplying z_0 by the sixth roots of unity, $w_6^k \;=\; e^{i\frac{k\pi}{3}}$ for $1\leq k\leq 5$. Therefore, the six 6th-roots of $5-i5\sqrt{3}$ are

$$10^{\frac{1}{6}}e^{i\frac{5\pi}{18}}, \quad 10^{\frac{1}{6}}e^{i\frac{11\pi}{18}}, \quad 10^{\frac{1}{6}}e^{i\frac{17\pi}{18}}, \quad 10^{\frac{1}{6}}e^{i\frac{23\pi}{18}}, \quad 10^{\frac{1}{6}}e^{i\frac{29\pi}{18}}, \quad \text{and} \quad 10^{\frac{1}{6}}e^{i\frac{35\pi}{18}}.$$

$\square$

Ex. 3. Using expressions for Re z and Im z, show that the hyperbola $x^2-y^2=1$ can be written as

$$z^2+\bar{z}^2=2.$$

SOLUTION . Recall that Re $(z)=\dfrac{z+\bar{z}}{2}$ and Im $(z)=\dfrac{z-\bar{z}}{2i}$. Therefore,

$$\begin{aligned}
x^2-y^2 &= (\mathrm{Re}(z))^2-(\mathrm{Im}(z))^2\\
&= \left(\frac{z+\bar{z}}{2}\right)^2-\left(\frac{z-\bar{z}}{2i}\right)^2\\
&= \frac{1}{4}\left(z^2+2z\bar{z}+\bar{z}^2+z^2-2z\bar{z}+\bar{z}^2\right)\\
&= \frac{1}{4}\left(2z^2+2\bar{z}^2\right)\\
&= \frac{1}{2}\left(z^2+\bar{z}^2\right).
\end{aligned}$$

Substituting this into the equation of the hyperbola, we arrive at

$$\frac{1}{2}\left(z^2+\bar{z}^2\right)=1,$$

in other words,

$$z^2 + \overline{z}^2 = 2.$$

□

Ex. 4. Find the four roots of the equation $z^4 + 4 = 0$ and use them to factor $z^4 + 4$ into quadratic factors with real coefficients.

SOLUTION . Rearranging the terms in the equation $z^4 + 4 = 0$ and expressing -4 in exponential form, we find that

$$z^4 \;=\; -4 \;=\; 4e^{i\pi}.$$

Therefore, the four roots of the equation are

$$z_0 = \sqrt{2}e^{i\frac{\pi}{4}}, \quad z_1 = \sqrt{2}e^{i\frac{3\pi}{4}}, \quad z_2 = \sqrt{2}e^{i\frac{5\pi}{4}}, \quad \text{and} \quad z_3 = \sqrt{2}e^{i\frac{7\pi}{4}}.$$

Note that z_0 and z_3 are conjugates of each other, while z_1 and z_2 are conjugates of each other. (These can be checked easily, e.g., $\overline{z_3} = \sqrt{2}e^{i(-\frac{7\pi}{4})} = \sqrt{2}e^{i\frac{\pi}{4}} = z_0$.) Notice that $re^{i\theta}$ and $re^{i(2\pi-\theta)}$ are conjugates.

Therefore,

$$\begin{aligned}
z^4 + 4 &= (z - z_0)(z - z_1)(z - z_2)(z - z_3) \\
&= [(z - z_0)(z - z_3)]\,[(z - z_1)(z - z_2)] \\
&= [(z - z_0)(z - \overline{z_0})]\,[(z - z_1)(z - \overline{z_1})] \\
&= \left[z^2 - (z_0 + \overline{z_0})z + z_0\overline{z_0}\right]\left[z^2 - (z_1 + \overline{z_1})z + z_1\overline{z_1}\right] \\
&= \left[z^2 - 2\mathrm{Re}(z_0)z + |z_0|^2\right]\left[z^2 - 2\mathrm{Re}(z_1)z + |z_1|^2\right] \\
&= \left[z^2 - 2\sqrt{2}\cos\left(\frac{\pi}{4}\right)z + \sqrt{2}^2\right]\left[z^2 + 2\sqrt{2}\cos\left(\frac{3\pi}{4}\right)z + \sqrt{2}^2\right] \\
&= \left[z^2 - 2z + 2\right]\left[z^2 + 2z + 2\right].
\end{aligned}$$

□